Lyapunov Stability Anlysis of a Non-Isothermal CSTR with LQR Control

Frank Vilimek

March 2026

1 Lyapunov Stability

A major component for designing a controller for any process, linear or nonlinear, is analyzing the stability of the controlled system. The strength of implementing controls comes when it is able to bring a system to a stable equilibrium point—this strength becomes a weakness when improper implementation makes the system unstable. For this reason, analyzing the system trajectories around equilibrium points must be done; for nonlinear systems, the method for analysis is through Lyapunov stability.

Lyapunov stability, a theory developed by Aleksandr Lyapunov, allows one to analyze the stability of equilibrium points for dynamical systems governed by nonlinear differential equations. Below we introduce the idea of Lyapunov stability, conditions for different kinds of stability, and criteria for making systems stable.

1.1 Stability Conditions

Before we can determine if a system is stable, we first need to define what "stable" means. To define stability, we consider a so-called "autonomous" nonlinear dynamical system (i.e., a nonlinear dynamical system whose derivatives are not explicitly with respect to the independent variable [1]),

$$\dot{x} = f(x(t)) \tag{1}$$

with an equilibrium point at the origin $x = 0$, making $f(0) = 0$. For this system, there are two main types of stability according to Lyapunov's theory [2]:

1. **Stability in the sense of Lyapunov**

   - Stability in the sense of Lyapunov requires that the system state is bounded within some "stability region" of radius ϵ. This stability condition means that a system can be dynamic and stable, so long as the state trajectories do not grow indefinitely. It may be considered a "weaker" requirement for stability, in the sense that the system does not need to reach one specific equilibrium state.
   - Imagine the classic example of a swinging pendulum with no friction or air resistance. The pendulum continues swinging, and it is impossible for its trajectories to get out of hand without some external disturbance.

- A Lyapunov stable equilibrium point requires that for every $\epsilon > 0$, there exists a $\delta > 0$ such that if $\|x(0)\| < \delta$ then for every $t \geq 0$ we have $\|x(t)\| < \epsilon$.

2. **Asymptotic stability**

   - Asymptotic stability requires that the system state approaches zero as $t \to \infty$. This type of stability is what may come to mind when one initially imagines a stable point. By definition, an asymptotically stable equilibrium must also be Lyapunov stable, so asymptotic stability is a "stronger" requirement for stability.
   - Imagine the concentration and temperature profiles for a CSTR under feedback control. The profiles all steadily reach an equilibrium point, and any over-/under-shoots decrease as $t \to \infty$.
   - An asymptotically stable equilibrium requires that there exists a $\delta > 0$ such that if $\|x(0)\| < \delta$, then $\lim_{t\to\infty} \|x(t)\| = 0$.

There is an additional type of stability, "exponential stability," which requires a known exponential rate at which the system stabilizes, however usually asymptotic stability is sufficient.

One could graphically show the stability of a system by creating a phase portrait, which allows you to visually see if a system is Lyapunov stable (if the states remain bounded over time) or asymptotically stable (if the states converge to a single point as $t \to \infty$).

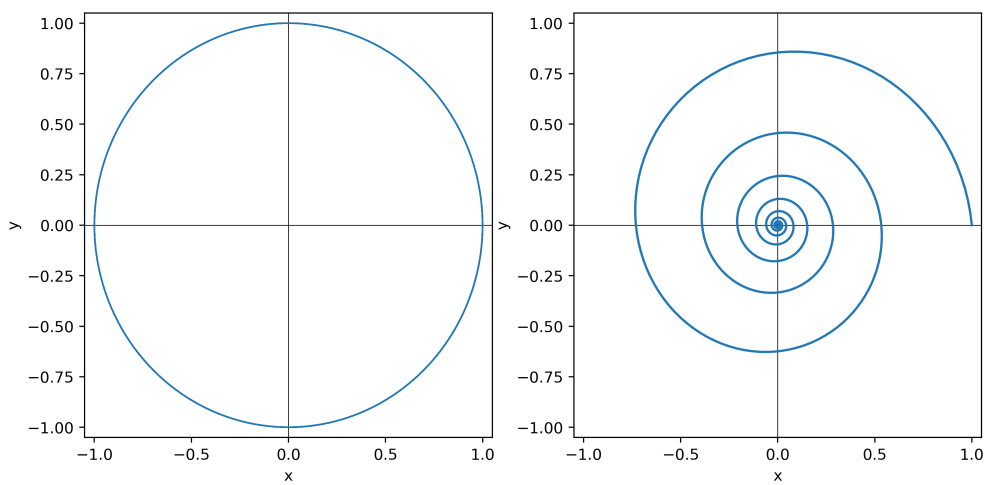

Figure 1: Example phase portraits for a system with Lyapunov stability (left) and a system with asymptotic stability (right). Both systems are evolving in time in the counter-clockwise direction, but this is purely model-dependent.

1.2 Lyapunov Functions and Lyapunov's Direct Method

Now that we know how stability is defined, we now introduce a way of determining stability for a system without requiring a closed-form solution. Consider the expression for kinetic energy,

$$KE = \frac{1}{2}mv^2 \tag{2}$$

If an object has zero velocity, then there is zero kinetic energy. If an object has some nonzero velocity, then it has nonzero kinetic energy—and no matter what speed that object has, it can never result in negative

energy. Think of kinetic energy as a result of being in a non-equilibrated state, where equilibrium is defined as zero velocity. If you are not at equilibrium, you would imagine that moving closer to equilibrium would mean that the kinetic energy decreases and vice versa.

All this was to introduce the idea of a Lyapunov function, which is often related to the "energy" of a system. There are technically infinitely-many Lyapunov functions $V(x)$ (each of them called a "Lyapunov function candidate"), and they all obey the following requirements [3]:

1. $V(x)$ is locally positive definite around $x = 0$ ($V(0) = 0$, $V(x) > 0$ on $\|x\| > 0$)
2. $V(x)$ is differentiable and its time derivative, $\dot{V}(x)$, is locally negative semi-definite ($\dot{V}(x) \leq 0$ on $\|x\| > 0$)

With this, it will be useful to introduce Lyapunov's theorem for local stability.

Lyapunov's Theorem for Stability

If there exists a Lyapunov function of the system $\dot{x} = f(x(t))$, then $x = 0$ is a stable equilibrium point. If in addition, $\dot{V}(x) < 0$, for $0 < \|x\| < r$ for some r, then $x = 0$ is asymptotically stable locally in the region $0 < \|x\| < r$.

If there exists a Lyapunov function that is radially unbounded ($V(x) \to \infty$ as $\|x\| \to \infty$) and $\dot{V}(x) < 0$ in the entire state space, then $x = 0$ is globally asymptotically stable.

Most often, a quadratic Lyapunov function of the form

$$V(x) = x^\top P x \tag{3}$$

with time derivative

$$\dot{V}(x) = x^\top (A^\top P + PA) x \tag{4}$$

is used. This choice is convenient because $V(x)$ is positive definite and $\dot{V}(x)$ is negative semi-definite when P is chosen such that

$$A^\top P + PA = -Q \tag{5}$$

where Q is a positive definite matrix, often chosen as the identity matrix I. Equation (5) is the Lyapunov equation, and if the solution P is positive definite then the system is asymptotically stable. Conveniently, Scipy has a function that solves this equation: `scipy.linalg.solve_continuous_lyapunov()`, which returns P. Keep in mind that the Scipy function solves the modified Lyapunov equation,

$$AP + PA^H = Q \tag{6}$$

which is very similar to the normal Lyapunov function, however the resulting solution is the negative of the one we want, so multiplying by -1 should be sufficient.

In our case, we do not necessarily need to confirm whether a Lyapunov function candidate is valid, we just need to solve the Lyapunov equation, so long as our model is linearized around an equilibrium point.

1.3 Example

Let us apply this to the (obviously stable) toy system described by,

$$\dot{x}_1 = -x_1$$
$$\dot{x}_2 = -x_2$$

Putting it into state-space representation gives

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

We can determine the stability in two ways: the first way is to choose a Lyapunov function candidate and determine if $\dot{V}(x) \leq 0$ on $\|x\| > 0$. The second way is to use the state matrix A in the Lyapunov equation, then see if the solution P has positive definite eigenvalues.

Let us start with the first method. Consider a Lyapunov function of the form $V(x_1, x_2) = x_1^2 + x_2^2$, which is positive definite and satisfies the first condition for using a Lyapunov function candidate. The time derivative of $V(x)$ is

$$\frac{dV}{dt} = \frac{dV}{dx}\frac{dx}{dt} = \frac{dV}{dx}\dot{x} = \begin{bmatrix} 2x_1 & 2x_2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Carrying out this vector-matrix multiplication yields

$$\frac{dV}{dt} = -2x_1^2 - 2x_2^2$$

which is negative semi-definite on $\|x\| > 0$, meaning this system is asymptotically stable. Let us now try the second method using the Lyapunov equation. Using the Scipy function, we obtain

$$P = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}$$

which has eigenvalues $\lambda_1 = \lambda_2 = 0.5$. P is thus a positive definite matrix, meaning the system is asymptotically stable.

1.4 Stability of an LTI System

Before we determine the stability of multivariate systems, it is educational to first analyze a simple single-state system, then extrapolate to multi-state systems. Consider the linear time-invariant system

$$\dot{x} = ax \tag{7}$$

where x is the state variable and a is a constant that depends on system dynamics. This is an ODE, which

when solved yields the evolution of the state in time (assuming $x(0) = 1$):

$$x(t) = e^{at} \tag{8}$$

We can see that the behavior of this system (i.e. whether it grows or decays in time) is dependent on the value of a—if a were positive, then this system is unbounded and, based on our definitions of stability from earlier, unstable; if a were negative, then this system would be asymptotically stable as the state is driven to the origin.

We can expand on the model in equation (7) by adding a control variable u and linear control law:

$$\begin{aligned} \dot{x} &= ax + bu \\ u &= -kx \end{aligned} \tag{9}$$

where b is a constant that depends on system dynamics and k is the constant controller gain. We can see that the control law $u = -kx$ can be directly substituted into the plant model to yield

$$\dot{x} = ax - bkx = (a - bk)x \tag{10}$$

Solving this ODE as before, we get the solution

$$x(t) = e^{(a-bk)t} \tag{11}$$

Now for this system to be stable, we require that $(a - bk) < 0$. This is straightforward to achieve: since a and b are determined by system dynamics and can't be changed, all we need to do is find the value of k that satisfies this. This idea can be extrapolated to a multi-state system described by the state-space representation of an LTI system with linear control

$$\begin{aligned} \dot{x} &= Ax + Bu \\ u &= -Kx \end{aligned} \tag{12}$$

where, for a stable system, we must choose a gain K such that $(A - BK) < 0$; in other words, we require that $(A - BK)$ is *Hurwitz*, meaning that the eigenvalues all have a strictly negative real part. This can be achieved in several ways: pole placement is a method where one chooses where the poles of the transfer function lie on the imaginary plane, and if all poles are on the left (negative) half then the system is stable; LQR is a method that, given a system is controllable, guarantees the gain matrix obtained is not only negative but also minimizes a quadratic cost function given penalty weights chosen by the control engineer.

2 CSTR Model

We can now use what we learned about stability to analyze the classic non-isothermal CSTR. As a refresher, a representation of the CSTR is below in figure (2).

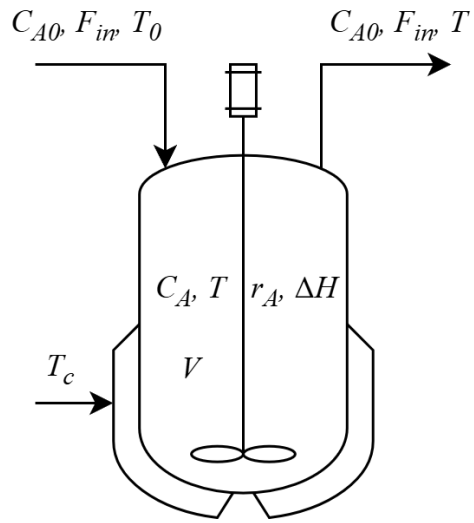

Figure 2: Non-isothermal CSTR with some relevant model parameters.

The mole and energy balance are described by the equations (13) and (14).

$$\frac{dC_A}{dt} = \frac{F_{in}}{V}(C_{A0} - C_A) - A_0 e^{-E/RT} C_A \tag{13}$$

$$\frac{dT}{dt} = \frac{F_{in}}{V}(T_0 - T) + \frac{(-\Delta H)}{\rho C_p V} A_0 e^{-E/RT} C_A - \frac{UA}{\rho C_p V}(T - T_c) \tag{14}$$

The temperature-dependent reaction is the first-order consumption of A to generate B: $A \to B$. This model is nonlinear, so before we can analyze the Lyapunov stability, we first need to linearize it around an equilibrium concentration C_A^{ss} and equilibrium temperature T^{ss}. We can borrow the expressions we found in the "LQR Applied to a Non-Isothermal CSTR" document I made previously:

$$C_A^{ss} = \frac{F_{in} C_{A0}}{F_{in} + V A_0 e^{-E/RT^{ss}}} \tag{15}$$

$$T^{ss} = \left[\frac{F_{in}}{V} + \frac{UA}{\rho C_p V}\right]^{-1} \left[\frac{F_{in}}{V} T_0 + \frac{UA}{\rho C_p V} T_c - \frac{F_{in} C_{A0} \Delta H e^{-E/RT^{ss}}}{\rho C_p \left(F_{in} + V A_0 e^{-E/RT^{ss}}\right)}\right] \tag{16}$$

We can then define the following deviation variables:

$$C_A' = C_A - C_A^{ss}, \qquad T' = T - T^{ss}, \qquad T_c' = T_c - T_c^{ss}. \tag{17}$$

Substitution of these deviation variables into equations (13) and (14) yields the linearized state-space model

$$\underbrace{\begin{bmatrix} \frac{dC_A'}{dt} \\ \frac{dT'}{dt} \end{bmatrix}}_{\dot{\mathbf{x}}} = \underbrace{\begin{bmatrix} -\left(\frac{F_{in}}{V} + k^{ss}\right) & \frac{Ek^{ss}}{RT^{ss2}} \\ -\frac{\Delta H k^{ss}}{\rho C_p} & \left(\frac{Ek^{ss}}{RT^{ss2}} - \frac{F_{in}}{V} - \frac{UA}{\rho C_p V}\right) \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} C_A' \\ T' \end{bmatrix}}_{\mathbf{x}} + \underbrace{\begin{bmatrix} 0 \\ \frac{UA}{\rho C_p V} \end{bmatrix}}_{B} \underbrace{T_c'}_{\mathbf{u}} \tag{18}$$

We can analyze the stability of the system without control by solving the Lyapunov function and seeing

if the real part of the eigenvalues of P are all strictly positive. Using the variable values we used in the LQR example, we have

$$P = \begin{bmatrix} 0.1758 & -0.2310 \\ -0.2310 & 3.8940 \end{bmatrix}$$

which has eigenvalues $\lambda_1 = 0.1615$ and $\lambda_2 = 3.9083$. Both of which are positive, meaning that technically our system is asymptotically stable without control, and this makes sense: the faster a reaction occurs, the more of A it is consuming. The more A it consumes, the lower the concentration of A, so the reaction works as its own regulator in a way. The system will eventually reach an equilibrium concentration and temperature on its own, based on the inlet flow rate, coolant temperature, etc.

Now let us investigate what happens when we introduce the linear control law. Doing so, instead of solving the Lyapunov equation with A defined solely as the state matrix, we will replace it with $(A - BK)$, which introduces the affect of the controller. Solving this Lyapunov equation yields

$$P = \begin{bmatrix} 0.1758 & -0.2245 \\ -0.2245 & 2.6156 \end{bmatrix}$$

with eigenvalues $\lambda_1 = 0.1554$ and $\lambda_2 = 2.6361$. Both eigenvalues are again positive, meaning that the controlled system in asymptotically stable as well. We should hope for this result, otherwise something went awry somewhere. We can also visually see the equilibrium point in the form of a phase portrait, seen in figure (3) below. This concludes this document.

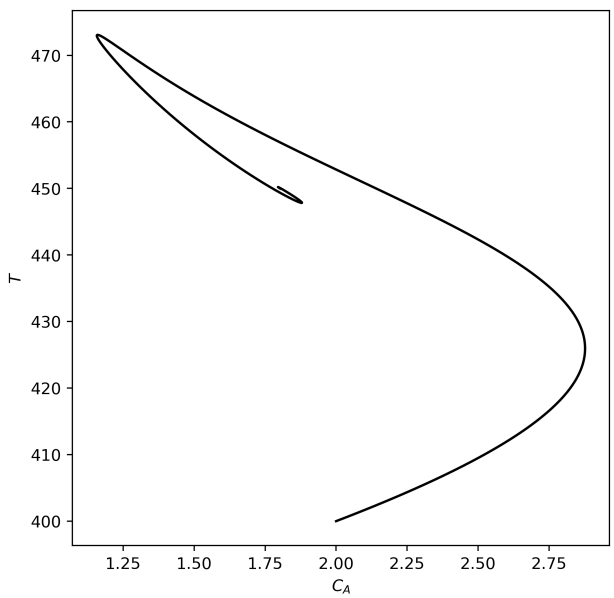

Figure 3: Phase portrait for the controlled non-isothermal CSTR. As expected, the states converge on the equilibrium point.

References

[1] Wikipedia, "Autonomous system (mathematics)." https://en.wikipedia.org/wiki/Autonomous_system_(mathematics). Date accessed: March 18, 2026.

[2] Wikipedia, "Lyapunov stability." https://en.wikipedia.org/wiki/Lyapunov_stability. Date accessed: March 18, 2026.

[3] Massachusetts Institute of Technology, "Lyapunov methods." https://dspace.mit.edu/bitstream/handle/1721.1/74611/6-241-fall-2003/contents/recitations/rec6.pdf, 2003. Date accessed: March 18, 2026.